
Nathan Sauldubois

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Summary —

I am a postdoctoral researcher at **New York University (NYU)**, working on probability, optimal transport, martingale optimal transport, and model risk. I obtained my **PhD in Applied Mathematics from École Polytechnique** under the supervision of **Nizar Touzi**, and defended my dissertation on **November 25, 2025**. My research combines theoretical probability, optimal control, optimal transport, and distributionally robust optimization.

Skills

Mathematical Interests: Optimal transport, martingale optimal transport, probability, optimal control, mean-field games, robustness

Numerical Interests: Machine learning, deep learning for PDEs, entropic regularization, particle methods
Programming: Python, R, OCaml, $\text{\LaTeX}$
Languages: French (native), English (fluent)

Publications

Published Papers & Preprints

- Sauldubois, N., and Touzi, N. (2024). *First-Order Martingale Model Risk and Semi-Static Hedging*. arXiv preprint arXiv:2410.06906. Under revision at *SIAM Journal on Financial Mathematics*.
- Compoin, A., Sauldubois, N., and Touzi, N. (2025). *Sensitivity Analysis in Stochastic Control*. arXiv preprint arXiv:2511.01828. Under revision at *Mathematical Finance*.
- Sauldubois, N. (2025). *Model Risk Static Hedging: A Constrained Distributionally Robust Optimization Approach*. arXiv preprint arXiv:2601.20139.
- Sauldubois, N., and Zhang, X. (2026). *Projected McKean–Vlasov Dynamics for Entropic Weak Optimal Transport*.

In Preparation

- Sauldubois, N. (2024). *Order- n Expansions in Classical and Causal Distributionally Robust Optimization*.
- Sauldubois, N., & Touzi, N. (2023). *A Linear–Quadratic Mean-Field Games Approach to the Marketplace Problem*.
- De March, H., Sauldubois, N., & Touzi, N. (2022). *Optimal Transport Meets Healthcare Optimization*.

Talks

- “A Linear–Quadratic Mean-Field Games Approach to the Marketplace Problem”, Séminaire Doctorant du CMAP (2023)
- “Extended L-Q Mean-Field Game with Common Noise”, Hammamet “Stochastic Control and Risk” (April 2023)
- “Robust Hedging and Distributionally Robust Sensitivity”, PhD Talk, Séminaire Bachelier (March 2024)
- “Robust Static Hedging and Distributionally Robust Sensitivity”, Hammamet Workshop, “Stochastic Control and Games for Risk and Regulation” (October 2024)
- “Model Risk and Semi-Static Hedging”, Bachelier Finance Society Meeting (July 2024)
- “Model Risk and Semi-Static Hedging”, Young Researchers Workshop, Berlin (2024)
- “Adapted Wasserstein Gradient Flow and Projected Langevin Dynamics for Weak Optimal Transport”, Hammamet “Stochastic Control and Risk” (April 2026)
- “Model Risk Static Hedging: A Constrained Distributionally Robust Optimization Approach”, Hammamet “Stochastic Control and Risk”; Bachelier Finance Society Meeting (July 2026)

Teaching

Exercise Sessions and Tutorials

- **Option Pricing & Stochastic Calculus** — Graduate course, M.S. in Financial Risk Engineering (FRE), NYU Tandon School of Engineering (Spring 2026)
- **Winter Math Bootcamp: From Brain Teasers to Black–Scholes** — Graduate bootcamp, M.S. in Financial Risk Engineering (FRE), NYU Tandon School of Engineering (Winter 2026)
- **Probability Refreshers** – Master X–HEC Data Science for Business (2022–2025)
- **Stochastic Models in Finance** – École Polytechnique (2022–2025)
- **Convex Optimization and Control** – Third-year bachelor’s course (2022–2025)
- **Preparatory Classes (CPGE)** – Oral exam preparation tutor at Lycée du Parc (2019–2020)

Supervision

- Co-supervision of the research internship of **Arthur Compoin** on *continuous-time model risk for stochastic optimization problems* (2025).

Administrative Responsibilities

- Responsible for communications and website maintenance for the **Séminaire Bachelier** (CMAP–CREST)
- Co-organizer of the **Séminaire Doctorant du CMAP**, 2023–2024

Education

École Polytechnique PhD in Applied Mathematics Defended on November 25, 2025. Supervisor: Nizar Touzi.	2022–2025
ENS de Lyon M2 Advanced Mathematics	2020–2021
ENS de Lyon M1 Advanced Mathematics	2019–2020
ENS de Lyon L3 Fundamental Mathematics with a Physics Minor	2018–2019

Research Experience

Postdoctoral Researcher, NYU – Distributionally robust optimization in continuous time. – Adapted Wasserstein gradient flows and applications to entropic martingale optimal transport and weak optimal transport.	2025–Present
CMAP, École Polytechnique PhD Research – Mean-field games with control-dependent interactions. – Sensitivity analysis and distributional robustness in stochastic control. – Martingale optimal transport and model risk.	2022–2025
Natixis – Quantitative Research Research Internship supervised by Pierre Henry-Labordère – Model uncertainty in fixed-income products. – Super-replication strategies for Bermudan swaptions with non-constant notionals. – Quantum optimal transport methods for correlation swaps.	2021–2022
CMAP, École Polytechnique Research Internship supervised by Nizar Touzi and Hadrien De March – Entropic optimal transport applied to healthcare policy design, in partnership with Quantev.	2021
Oxford University Research Internship supervised by Julien Berestycki – Analysis of high-dimensional branching Brownian motion and extremal particle estimates.	2020
Université de Bordeaux Research Internship supervised by David Lannes – Study of shallow-water wave equations and the behavior of solutions in bounded-depth domains.	2019

Additional Activities

- Special Jury Prize – SMF Junior Contest (2018)